

Quantum Rabi Oscillations, Collapse, and Revival in the Jaynes–Cummings Model

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Abstract

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photon-number statistics of the initial field. For a Fock state, the atomic population undergoes sinusoidal quantum Rabi oscillations with a photon-number-dependent frequency. For a coherent state, the superposition of several photon-number components produces the collapse and subsequent revival of the atomic oscillations.

The model is obtained by restricting the atomic Hilbert space to two energy levels, considering a single field mode, and applying the rotating-wave approximation. Under these assumptions, the total excitation number is conserved, and the Hilbert space separates into independent two-dimensional manifolds. This structure allows the Hamiltonian to be diagonalized analytically in terms of atom–field dressed states.

In this work, the excitation manifolds and dressed-state spectrum are derived, followed by the calculation of the atomic excitation probability for arbitrary initial photon-number statistics. The cases of vacuum, Fock, and coherent initial fields are studied separately. Particular attention is given to quantum Rabi oscillations and to the physical origin and characteristic time scales of collapse and revival.

Introduction

The Jaynes–Cummings model describes the interaction between a two-level atom and a single quantized mode of the electromagnetic field. It is one of the fundamental exactly solvable models of quantum optics and provides a simple framework for studying the coherent exchange of excitations between matter and radiation.

Unlike the semiclassical Rabi model, in which the electromagnetic field is described by a prescribed classical function, the Jaynes–Cummings model treats both the atom and the field quantum mechanically. Consequently, the atomic dynamics depend on the

Jaynes–Cummings Hamiltonian

The total Hamiltonian of a two-level atom interacting with a single quantized field mode is

$$\hat{H}_{\text{JC}} = \frac{\hbar\omega_a}{2}\hat{\sigma}_z + \hbar\omega_c\hat{a}^\dagger\hat{a} + i\hbar g\left(\hat{a}^\dagger\hat{\sigma}_- - \hat{a}\hat{\sigma}_+\right). \quad (1)$$

Here, ω_a is the atomic transition frequency, ω_c is the cavity-mode frequency, and g is the atom–field coupling strength. The operators $\hat{a}$ and $\hat{a}^\dagger$ annihilate and create one photon in the cavity mode, respectively, while $\hat{\sigma}_+$ and $\hat{\sigma}_-$ raise and lower the atomic state.

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The field zero-point energy, $\hbar\omega_c/2$, has been omitted because it produces only a constant shift of the total energy.

Equation (1) is obtained after applying the rotating-wave approximation. The retained interaction terms describe excitation exchange:

$$|g, n\rangle \longleftrightarrow |e, n-1\rangle. \quad (2)$$

The term $\hat{a}\hat{\sigma}_+$ corresponds to photon absorption and atomic excitation, whereas $\hat{a}^\dagger\hat{\sigma}_-$ describes photon emission and atomic de-excitation.

Conservation of the excitation number

The Jaynes–Cummings Hamiltonian conserves the total excitation-number operator

$$\hat{N} = \hat{a}^\dagger\hat{a} + \hat{\sigma}_+\hat{\sigma}_-. \quad (3)$$

Indeed,

$$[\hat{H}_{\text{JC}}, \hat{N}] = 0. \quad (4)$$

Therefore, the Hilbert space separates into independent excitation manifolds. For each integer $n \geq 1$,

$$\mathcal{H}_n = \text{span}\{|n, g\rangle, |n-1, e\rangle\}. \quad (5)$$

The state $|n, g\rangle$ contains n photons and an atom in the ground state, whereas $|n-1, e\rangle$ contains $n-1$ photons and an excited atom. Both states have the same total number of excitations.

In the ordered basis

$$\{|n, g\rangle, |n-1, e\rangle\}, \quad (6)$$

the Hamiltonian restricted to $\mathcal{H}_n$ is

$$\hat{H}^{(n)} = \hbar \begin{pmatrix} n\omega_c - \omega_a/2 & ig\sqrt{n} \\ -ig\sqrt{n} & (n-1)\omega_c + \omega_a/2 \end{pmatrix}. \quad (7)$$

The state $|0, g\rangle$ belongs to a one-dimensional manifold and is an eigenstate of the Hamiltonian because it contains no excitation that can be exchanged.

Energy spectrum and dressed states

We define the atom–cavity detuning as

$$\Delta = \omega_c - \omega_a. \quad (8)$$

Diagonalizing Eq. (7), the eigenenergies are

$$E_{\pm}^{(n)} = \hbar\omega_c \left(n - \frac{1}{2} \right) \pm \frac{\hbar\Omega_n}{2}, \quad (9)$$

where

$$\Omega_n = \sqrt{\Delta^2 + 4ng^2} \quad (10)$$

is the generalized quantum Rabi frequency.

The corresponding eigenstates may be written as

$$|\psi_+^{(n)}\rangle = \cos\theta_n |n, g\rangle - i \sin\theta_n |n-1, e\rangle, \quad (11)$$

$$|\psi_-^{(n)}\rangle = \sin\theta_n |n, g\rangle + i \cos\theta_n |n-1, e\rangle, \quad (12)$$

with

$$\cos\theta_n = \sqrt{\frac{\Omega_n - \Delta}{2\Omega_n}}, \quad \sin\theta_n = \sqrt{\frac{\Omega_n + \Delta}{2\Omega_n}}. \quad (13)$$

Together with the state $|0, g\rangle$, these atom–field eigenstates are known as dressed states.

At resonance, $\Delta = 0$, and therefore

$$\theta_n = \frac{\pi}{4}. \quad (14)$$

The dressed states then become equal superpositions of the bare atom–field states:

$$|\psi_+^{(n)}\rangle = \frac{1}{\sqrt{2}} (|n, g\rangle - i |n-1, e\rangle), \quad (15)$$

$$|\psi_-^{(n)}\rangle = \frac{1}{\sqrt{2}} (|n, g\rangle + i |n-1, e\rangle). \quad (16)$$

The energy splitting at resonance is

$$E_+^{(n)} - E_-^{(n)} = 2\hbar g\sqrt{n}. \quad (17)$$

Atomic dynamics for an arbitrary initial field

Let the atom initially be in its ground state and the field in an arbitrary pure state,

$$|\Psi(0)\rangle = \sum_{n=0}^{\infty} c_n |n, g\rangle. \quad (18)$$

Using the dressed-state basis,

$$|\Psi(0)\rangle = c_0 |0, g\rangle + \sum_{n=1}^{\infty} c_n \left(\cos\theta_n |\psi_+^{(n)}\rangle + \sin\theta_n |\psi_-^{(n)}\rangle \right). \quad (19)$$

The dressed states evolve according to

$$|\psi_{\pm}^{(n)}\rangle \longrightarrow e^{-iE_{\pm}^{(n)}t/\hbar} |\psi_{\pm}^{(n)}\rangle. \quad (20)$$

The state $|0, g\rangle$ evolves only through a phase,

$$|0, g\rangle \longrightarrow e^{i\omega_a t/2} |0, g\rangle. \quad (21)$$

The complete state at time t is therefore

$$\begin{aligned} |\Psi(t)\rangle = & e^{i\omega_a t/2} c_0 |0, g\rangle \\ & + \sum_{n=1}^{\infty} c_n e^{-i(n-1/2)\omega_c t} \left[e^{-i\Omega_n t/2} \cos \theta_n \left| \psi_+^{(n)} \right\rangle \right. \\ & \left. + e^{i\Omega_n t/2} \sin \theta_n \left| \psi_-^{(n)} \right\rangle \right]. \end{aligned} \quad (22)$$

The probability of finding the atom in the excited state is

$$P_e(t) = \sum_{n=1}^{\infty} |c_n|^2 \frac{4ng^2}{\Omega_n^2} \sin^2 \left(\frac{\Omega_n t}{2} \right). \quad (23)$$

Equivalently,

$$P_e(t) = \sum_{n=1}^{\infty} |c_n|^2 \frac{1 - \cos(\Omega_n t)}{2[1 + \Delta^2/(4ng^2)]}. \quad (24)$$

Each photon-number component generates a Rabi oscillation with a different frequency and statistical weight.

Quantum Rabi oscillations

Initial vacuum field

Consider first the absolute ground state,

$$|\Psi(0)\rangle = |0, g\rangle. \quad (25)$$

In this case,

$$c_n = \delta_{n0}, \quad (26)$$

and therefore

$$P_e(t) = 0. \quad (27)$$

The atom remains in the ground state because the system initially contains no excitation that can be transferred from the field to the atom.

Initial Fock state

Suppose that the cavity field is initially prepared in a Fock state with N photons:

$$|\Psi(0)\rangle = |N, g\rangle. \quad (28)$$

The field coefficients are

$$c_n = \delta_{nN}. \quad (29)$$

Equation (23) then reduces to

$$P_e^{(N)}(t) = \frac{4Ng^2}{\Omega_N^2} \sin^2 \left(\frac{\Omega_N t}{2} \right), \quad (30)$$

where

$$\Omega_N = \sqrt{\Delta^2 + 4Ng^2}. \quad (31)$$

At resonance,

$$P_e^{(N)}(t) = \sin^2 \left(g\sqrt{N} t \right). \quad (32)$$

Thus, the atom oscillates between the states $|N, g\rangle$ and $|N-1, e\rangle$. These population oscillations are known as quantum Rabi oscillations. Their frequency increases as $\sqrt{N}$, reflecting the bosonic enhancement of the atom–field interaction.

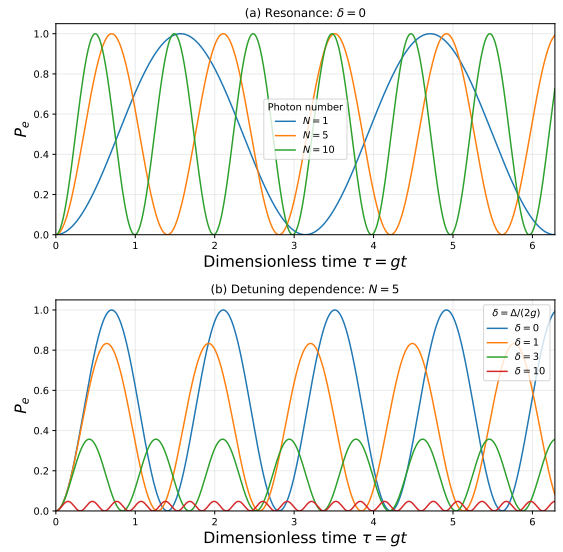


Figure 1: Quantum Rabi oscillations for an atom initially in the ground state and a field prepared in a Fock state. (a) Excited-state probability at resonance for different photon numbers. The oscillation frequency increases as $\Omega_N = 2g\sqrt{N}$. (b) Excited-state probability for fixed $N = 5$ and different relative detunings $\delta = \Delta/(2g)$. Increasing the detuning reduces the maximum excitation probability and modifies the oscillation frequency.

Collapse and revival for an initial coherent field

Coherent states

A coherent field state is defined as

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (33)$$

Its mean photon number is

$$\bar{n} = |\alpha|^2, \quad (34)$$

and its photon-number distribution is Poissonian:

$$P_n = |c_n|^2 = e^{-\bar{n}} \frac{\bar{n}^n}{n!}. \quad (35)$$

Coherent states reproduce several properties of a classical monochromatic field. They have a well-defined phase, minimize the field-quadrature uncertainty, and possess equal photon-number mean and variance:

$$\text{Var}(n) = \bar{n}. \quad (36)$$

Therefore, the relative photon-number uncertainty is

$$\frac{\sqrt{\text{Var}(n)}}{\bar{n}} = \frac{1}{\sqrt{\bar{n}}}, \quad (37)$$

which decreases in the large-photon-number limit.

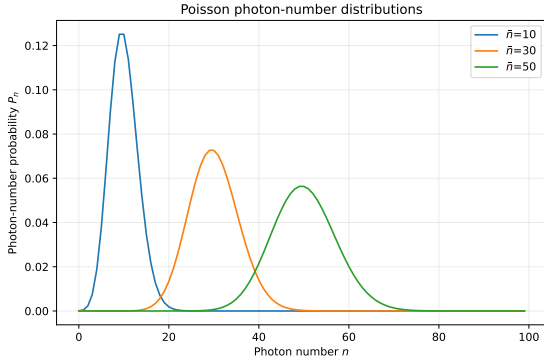


Figure 2: Poisson photon-number distributions $P_n = e^{-\bar{n}} \bar{n}^n / n!$ for coherent states with different mean photon numbers. The vertical dotted lines indicate the corresponding mean values. The absolute width increases approximately as $\sqrt{\bar{n}}$, whereas the relative width decreases as $1/\sqrt{\bar{n}}$.

Collapse of the Rabi oscillations

For simplicity, consider the resonant case,

$$\Delta = 0. \quad (38)$$

Using Eq. (35), the excited-state probability is

$$P_e(t) = \frac{1}{2} - \frac{1}{2} \sum_{n=0}^{\infty} e^{-\bar{n}} \frac{\bar{n}^n}{n!} \cos(2g\sqrt{n}t). \quad (39)$$

Each excitation manifold contributes a Rabi oscillation with frequency

$$\Omega_n = 2g\sqrt{n} \quad (40)$$

and weight determined by the Poisson distribution. Because the frequencies depend on photon number, the different terms progressively acquire different phases.

In the large-photon-number limit,

$$\bar{n} \gg 1, \quad (41)$$

the Poisson distribution can be approximated by a Gaussian:

$$e^{-\bar{n}} \frac{\bar{n}^n}{n!} \approx \frac{1}{\sqrt{2\pi\bar{n}}} \exp\left[-\frac{(n-\bar{n})^2}{2\bar{n}}\right]. \quad (42)$$

Expanding $\sqrt{n}$ around $\bar{n}$ gives

$$\sqrt{n} \approx \sqrt{\bar{n}} + \frac{n-\bar{n}}{2\sqrt{\bar{n}}}. \quad (43)$$

Using

$$\cos(2g\sqrt{n}t) = \text{Re}\left[e^{2ig\sqrt{n}t}\right], \quad (44)$$

the sum may be approximated by the Gaussian integral

$$P_e(t) \approx \frac{1}{2} - \frac{1}{2} \text{Re} \left[\int_{-\infty}^{+\infty} \frac{dn}{\sqrt{2\pi\bar{n}}} \exp\left(-\frac{(n-\bar{n})^2}{2\bar{n}}\right) \times \exp\left(2ig\left[\sqrt{\bar{n}} + \frac{n-\bar{n}}{2\sqrt{\bar{n}}}\right]t\right) \right]. \quad (45)$$

Evaluating the integral gives

$$P_e(t) \approx \frac{1}{2} - \frac{1}{2} e^{-g^2 t^2 / 2} \cos(2g\sqrt{\bar{n}}t). \quad (46)$$

The central oscillation frequency,

$$\Omega_{\bar{n}} = 2g\sqrt{\bar{n}}, \quad (47)$$

coincides with the semiclassical Rabi frequency associated with the mean field amplitude. However, the oscillations are modulated by the Gaussian envelope

$$e^{-g^2 t^2 / 2}. \quad (48)$$

It results from dephasing among the different photon-number components of the coherent state.

A collapse time may be defined by requiring that the envelope decrease to e^{-1} :

$$e^{-g^2 t_c^2 / 2} = e^{-1}. \quad (49)$$

This gives

$$t_c = \frac{\sqrt{2}}{g}. \quad (50)$$

In the large-photon-number regime,

$$\frac{\Omega_{\bar{n}}}{1/t_c} \sim \sqrt{\bar{n}} \gg 1, \quad (51)$$

so several approximately semiclassical Rabi cycles occur before the collapse.

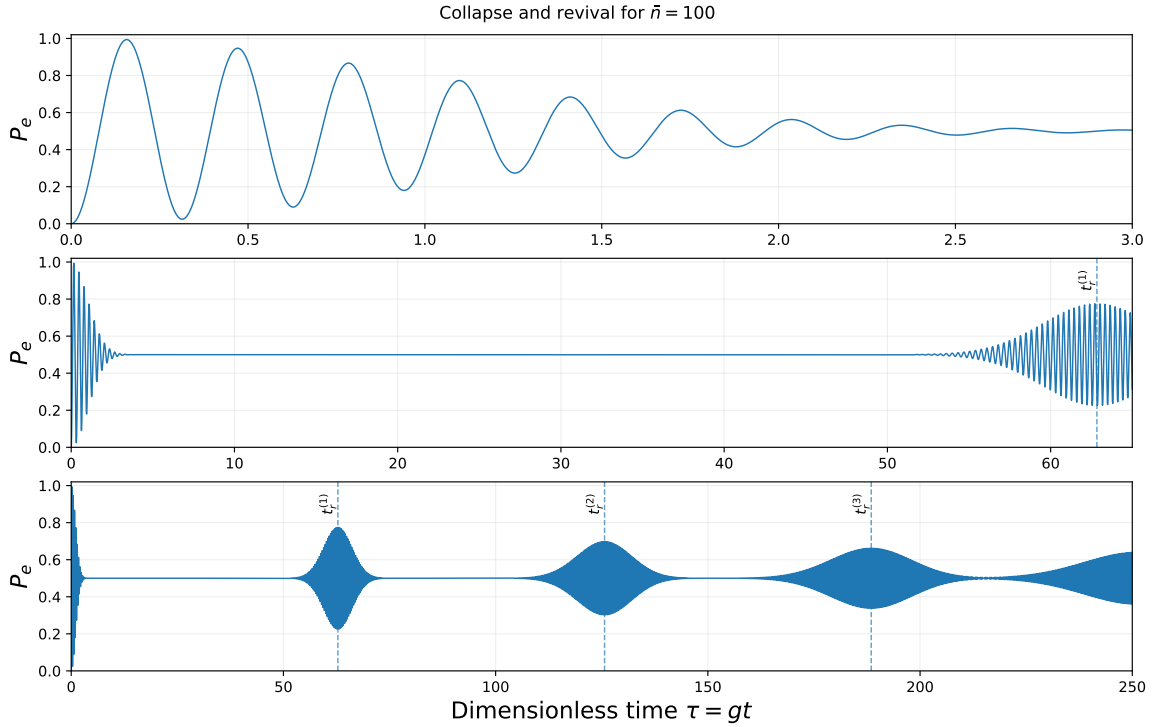


Figure 3: Excited-state probability for an atom initially in the ground state and a resonant coherent field with mean photon number $\bar{n} = 100$. The upper panel shows the initial quantum Rabi oscillations and their collapse. The middle panel shows the first revival near $gt_r^{(1)} = 2\pi\sqrt{\bar{n}} \approx 62.8$. The lower panel displays several successive revivals. Vertical dashed lines indicate the analytical predictions $gt_r^{(m)} = 2\pi m\sqrt{\bar{n}}$.

Revival of the Rabi oscillations

Initially, the dominant photon-number components oscillate approximately in phase with frequencies close to

$$\Omega_{\bar{n}} = 2g\sqrt{\bar{n}}. \quad (52)$$

As time increases, the different components dephase and their sum approaches its average value. At later times, however, the discrete frequency components can rephase, producing a revival of the Rabi oscillations.

For neighboring photon numbers,

$$\Omega_{n+1} - \Omega_n = 2g(\sqrt{n+1} - \sqrt{n}). \quad (53)$$

For $n \approx \bar{n} \gg 1$,

$$\Omega_{n+1} - \Omega_n \approx \frac{g}{\sqrt{\bar{n}}}. \quad (54)$$

The components rephase when their phase difference becomes an integer multiple of 2π :

$$(\Omega_{\bar{n}+1} - \Omega_{\bar{n}}) t_r^{(m)} = 2\pi m, \quad m \in \mathbb{N}. \quad (55)$$

Therefore, the revival times are approximately

$$t_r^{(m)} \approx \frac{2\pi m\sqrt{\bar{n}}}{g}, \quad m \in \mathbb{N}. \quad (56)$$

In dimensionless units,

$$gt_r^{(m)} \approx 2\pi m\sqrt{\bar{n}}. \quad (57)$$

The collapse and revival phenomenon is a consequence of the discrete photon-number distribution of the quantized field and has no direct counterpart in a model driven by a single classical Rabi frequency.

Discussion

The Jaynes–Cummings dynamics depend strongly on the initial state of the field. For an initial vacuum field and an atom in its ground state, no population transfer occurs because the system contains no excitation. For a Fock state, only one excitation manifold contributes, producing a single sinusoidal oscillation with frequency proportional to $\sqrt{N}$.

For a coherent state, several excitation manifolds contribute simultaneously. The initial dynamics resemble semiclassical Rabi oscillations because the Poisson distribution is concentrated around its mean photon number. The finite width of the distribution, however, gives rise to a range of quantum Rabi frequencies. Their dephasing produces the collapse,

while their subsequent rephasing produces the revivals.

The collapse envelope and revival time depend differently on the mean photon number. To leading order, the collapse time is of order $1/g$ and is approximately independent of $\bar{n}$. In contrast, the first revival time increases as $\sqrt{\bar{n}}/g$. Consequently, the separation between the collapse and the first revival becomes larger as the coherent field becomes more intense.

Conclusions

The interaction between a two-level atom and a single quantized electromagnetic mode was studied within the Jaynes–Cummings model. After applying the rotating-wave approximation, the total excitation number is conserved, allowing the Hilbert space to be decomposed into independent two-dimensional manifolds.

Diagonalization of each manifold gives the atom–field dressed states and the generalized quantum Rabi frequency

$$\Omega_n = \sqrt{\Delta^2 + 4ng^2}. \quad (58)$$

For an initial Fock state, the atom undergoes periodic quantum Rabi oscillations whose frequency scales as $\sqrt{N}$. An initial vacuum field cannot excite an atom prepared in its ground state because no excitation is available for exchange.

For an initial coherent field, the different photon-number components oscillate at different Rabi frequencies. Their progressive dephasing produces the collapse of the atomic oscillations, while their later rephasing produces revivals at approximately

$$t_r^{(m)} = \frac{2\pi m \sqrt{\bar{n}}}{g}. \quad (59)$$

These results demonstrate how the quantum statistics of the field determine the atomic dynamics. In particular, collapse and revival provide a direct signature of the discrete photon-number distribution and of the correlations generated between the atom and the cavity mode.